

Sigma-ML

Wireless Network Measurement and Field Testing Solution

Technical Product Documentation

1. Introduction

Overview

Sigma-ML is a compact, lightweight wireless network measurement application developed for smartphones. It enables network engineers, telecom operators, service providers, and field technicians to perform comprehensive wireless network testing without requiring bulky laptops or dedicated measurement hardware.

The application transforms an Android smartphone into a professional wireless network testing device capable of collecting detailed network measurements from the physical layer (Layer 1) to the application layer. By automating repetitive testing tasks and providing real-time protocol analysis, Sigma-ML significantly improves the efficiency of wireless network deployment, optimization, troubleshooting, and maintenance.

Unlike traditional drive-test equipment that requires expensive hardware and complicated configurations, Sigma-ML operates directly on commercially available smartphones. This portability makes it suitable for outdoor drive testing, indoor surveys, network optimization projects, acceptance testing, and long-term unattended monitoring.

The software supports multiple generations of cellular technologies, including 5G NR, LTE, LTE-Advanced, WCDMA, HSPA, EDGE, and GPRS. It also supports Wi-Fi measurements, enabling engineers to evaluate heterogeneous wireless environments using a single application.

Sigma-ML is designed to simplify wireless network performance analysis while maintaining professional-grade measurement accuracy. The application collects radio parameters, protocol messages, throughput statistics, voice quality metrics, GPS information, and application-level performance data, enabling engineers to identify network issues and determine their root causes efficiently.

2. Purpose of Sigma-ML

Wireless communication networks are becoming increasingly complex due to the deployment of advanced technologies such as 5G, carrier aggregation, VoLTE, and heterogeneous wireless environments. Network operators require reliable measurement tools to monitor network quality, evaluate user experience, and troubleshoot performance issues.

Sigma-ML addresses these requirements by providing a smartphone-based measurement platform that automates network testing while collecting detailed performance data.

The primary objectives of Sigma-ML include:

- Measuring wireless network quality.
- Evaluating cellular network coverage.

- Monitoring throughput and latency.
- Recording detailed radio measurements.
- Performing automated field testing.
- Measuring voice quality.
- Supporting indoor and outdoor surveys.
- Identifying network performance bottlenecks.
- Simplifying root cause analysis.
- Reducing operational costs associated with drive testing.

By automating complex testing procedures, Sigma-ML minimizes manual intervention and improves the consistency of measurement results.

3. Product Objectives

Sigma-ML has been designed with several key objectives:

Improve Network Performance

Provide accurate measurements that help engineers optimize cellular network coverage and performance.

Simplify Field Testing

Allow engineers to perform professional drive tests using only smartphones.

Reduce Testing Cost

Eliminate the need for expensive measurement equipment while maintaining reliable results.

Automate Measurements

Enable unattended testing through scheduling and automatic execution of predefined test scenarios.

Support Multiple Technologies

Provide comprehensive measurement capabilities across multiple generations of wireless communication systems.

Enhance Troubleshooting

Collect protocol-level information that assists engineers in identifying network problems efficiently.

4. System Architecture

Sigma-ML follows a smartphone-centric architecture where the mobile device acts as the primary measurement platform.

5. Working Principle

Sigma-ML continuously communicates with the smartphone modem and operating system to retrieve radio information, protocol messages, application performance statistics, and location data.

The working process consists of the following stages:

Step 1 – Test Configuration

The engineer configures the required measurements, including:

- FTP Download
 - FTP Upload
 - HTTP Download
 - Voice Calls
 - SMS
 - Ping Tests
 - Automatic Scheduling
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Step 2 – Measurement Execution

The application performs all configured tests automatically according to predefined schedules.

Examples include:

- Automatic file download
 - Automatic upload
 - Periodic ping transmission
 - Voice call origination
 - Incoming call reception
 - SMS transmission
 - Data session monitoring
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Step 3 – Data Collection

During execution, Sigma-ML continuously collects:

- Signal strength
- Cell information
- Protocol messages
- GPS coordinates

- Throughput
 - Delay
 - Packet loss
 - Voice quality
 - Device status
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Step 4 – Data Logging

The collected information is stored locally within the smartphone.

The application supports:

- Internal storage
 - External memory cards
 - Continuous logging
 - Long-duration recording
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Step 5 – Data Export

After completion, engineers can export collected measurements into:

- CSV
- DLF

These files can be analyzed using external reporting and visualization tools.

6. Key Characteristics

Sigma-ML provides several characteristics that distinguish it from conventional drive-test systems.

Smartphone-Based Measurement

The application runs directly on supported Android smartphones, eliminating the need for laptops and specialized equipment.

Lightweight Deployment

Installation requires only the application and supported permissions.

No external receiver or measurement hardware is necessary.

Portable Solution

Field engineers can carry a single smartphone instead of transporting multiple hardware devices. This greatly improves operational efficiency during large-scale measurement campaigns.

Real-Time Monitoring

Sigma-ML displays collected measurements instantly.

Engineers can observe:

- Signal strength
- Serving cell
- Neighbor cells
- Throughput
- GPS position
- Voice quality
- Protocol messages

This enables immediate identification of network issues.

Automated Testing

One of Sigma-ML's major strengths is automation.

Users define test scenarios once, and Sigma-ML executes them automatically.

Automation includes:

- FTP download
- FTP upload
- HTTP download
- Voice calls
- SMS tests
- Ping
- File transfers
- Timed execution

This reduces human error and enables unattended testing.

7. Benefits of Smartphone-Based Measurements

Traditional network measurement solutions often require:

- Laptop computers
- Dedicated drive-test equipment
- External scanners
- Multiple cables
- Power supplies
- GPS receivers

Sigma-ML simplifies this setup by integrating all measurement capabilities into a smartphone.

Benefits include:

- Reduced equipment cost
- Faster deployment
- Increased mobility
- Simplified operation
- Lower maintenance requirements
- Improved portability
- Rapid field testing
- Easier training for engineers

Organizations can therefore conduct frequent measurement campaigns without investing in expensive testing infrastructure.

8. Typical Users

Sigma-ML is designed for a wide range of professionals involved in wireless communication networks.

Typical users include:

- Mobile Network Operators
- Telecom Service Providers
- RF Engineers
- Drive Test Engineers
- Network Optimization Teams
- Quality Assurance Engineers
- Infrastructure Vendors
- Research Organizations
- Universities

- Wireless Technology Consultants
- Field Support Engineers

Each user group benefits from Sigma-ML's ability to collect accurate wireless measurements while minimizing operational complexity.

9. Core Features of Sigma-ML

Sigma-ML provides a comprehensive set of features that enable engineers to monitor, measure, and analyze wireless network performance using a smartphone. These features are designed to simplify field testing while providing detailed insights into network behavior across different communication layers.

9.1 Automatic Test Execution

One of the primary strengths of Sigma-ML is its ability to automate repetitive testing tasks. Instead of manually initiating each test, users can configure a testing profile and allow the application to execute the required operations automatically.

Supported automated tasks include:

- FTP Download (DL)
- FTP Upload (UL)
- HTTP Download
- Ping Test
- Voice Call Origination
- Voice Call Termination
- SMS Sending and Receiving
- Continuous Data Session Monitoring
- Scheduled Test Execution

Automation reduces manual effort, improves test consistency, and enables long-duration monitoring without continuous user interaction.

9.2 Real-Time Monitoring

Sigma-ML continuously displays live network information while measurements are in progress. Engineers can monitor changing radio conditions and identify network issues immediately.

Typical real-time information includes:

- Signal strength
- Serving cell information
- Neighboring cells

- Frequency band
- Cell Identifier (Cell ID)
- Physical Cell Identity (PCI)
- Channel bandwidth
- Data throughput
- Voice call status
- GPS location
- Device battery level
- Storage utilization

Real-time visualization allows engineers to observe network behavior during movement, making it particularly useful for drive testing and coverage verification.

9.3 User Event Marker

Sigma-ML includes a **User Event Marker** feature that enables engineers to manually mark significant events during testing.

Examples include:

- Signal drop observed
- Call disconnected
- Network handover
- Coverage hole detected
- Building entry
- Tunnel crossing
- Customer complaint location

Each marker is stored along with the corresponding measurement data and timestamp, making it easier to locate important events during post-processing and analysis.

9.4 Indoor and Outdoor Measurement

Sigma-ML supports both outdoor and indoor testing environments.

Outdoor Measurement

Outdoor testing uses GPS to record the geographical location of each measurement point. This is particularly useful for:

- Drive testing

- Coverage verification
- Route optimization
- Mobility testing
- Handover analysis

Indoor Measurement

Since GPS signals may be unavailable indoors, Sigma-ML allows users to place manual location markers. Engineers can assign meaningful labels such as:

- Floor 1
- Conference Room
- Basement
- Server Room
- Shopping Mall Entrance

These markers help organize indoor measurements and simplify coverage analysis.

10. Measurement Capabilities

Sigma-ML captures information from multiple layers of the communication stack, allowing engineers to analyze wireless network performance from both radio and application perspectives.

10.1 Layer 1 (Physical Layer) Measurements

Layer 1 focuses on radio signal quality and physical transmission characteristics.

Typical Layer 1 measurements include:

- RSSI (Received Signal Strength Indicator)
- RSRP (Reference Signal Received Power)
- RSRQ (Reference Signal Received Quality)
- SINR (Signal to Interference plus Noise Ratio)
- CQI (Channel Quality Indicator)
- Frequency Band
- EARFCN / NRARFCN
- Transmission Power
- Modulation Information
- Carrier Aggregation Status

These measurements help determine the strength and quality of the wireless signal received by the device.

10.2 Layer 2 (Data Link Layer)

Layer 2 measurements provide information regarding data transmission efficiency between the device and the network.

Examples include:

- MAC Layer Statistics
- HARQ Retransmissions
- Buffer Status
- Scheduling Information
- Packet Transmission Status
- Data Queue Information
- Error Recovery Statistics

Layer 2 analysis assists engineers in understanding transmission delays and identifying retransmission-related issues.

10.3 Layer 3 (Network Layer)

Layer 3 contains signaling messages exchanged between the smartphone and the mobile network.

Sigma-ML provides real-time decoding of Layer 3 protocol messages, enabling detailed network analysis.

Examples include:

- Cell Selection
- Cell Reselection
- Handover Messages
- Attach Procedures
- Registration Procedures
- Authentication
- Tracking Area Update (TAU)
- Radio Resource Control (RRC)
- Mobility Management
- Session Management

These protocol messages are essential for identifying signaling failures and understanding device behavior during mobility.

10.4 Application Layer Measurements

In addition to radio information, Sigma-ML measures application-level performance.

Examples include:

- FTP Download Speed
- FTP Upload Speed
- HTTP Download Time
- Web Response Time
- File Transfer Success Rate
- Data Session Duration
- Network Availability

These measurements represent the actual user experience and help correlate network conditions with application performance.

11. Supported Test Types

Sigma-ML supports multiple testing methodologies used by telecom operators and network optimization teams.

FTP Download Test

Measures download throughput from a remote FTP server.

Collected parameters include:

- Download Speed
 - Transfer Duration
 - Packet Loss
 - Network Stability
 - Average Throughput
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FTP Upload Test

Measures upload performance by transmitting files to an FTP server.

Typical observations include:

- Upload Speed

- Upload Completion Time
 - Network Interruptions
 - Data Transmission Errors
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HTTP Download Test

Evaluates application performance by downloading content using the HTTP protocol.

Measurements include:

- Response Time
 - Download Rate
 - Content Retrieval Time
 - Connection Success Rate
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Ping Test

Ping testing evaluates network latency and connectivity.

Typical parameters include:

- Round Trip Time (RTT)
 - Packet Loss
 - Average Delay
 - Maximum Delay
 - Minimum Delay
 - Network Availability
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SMS Test

Sigma-ML can automatically send and receive SMS messages to verify messaging services.

Collected metrics include:

- Delivery Time
 - Success Rate
 - Failure Count
 - Retry Attempts
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Voice Call Test

Voice call testing measures call setup and call maintenance performance.

Measurements include:

- Call Setup Time
 - Call Success Rate
 - Call Drop Rate
 - Call Duration
 - Call Completion Rate
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12. Supported Wireless Technologies

Sigma-ML supports multiple generations of wireless communication technologies, making it suitable for testing modern heterogeneous networks.

5G NR (New Radio)

Supports measurement of:

- Signal Quality
 - Coverage
 - Throughput
 - Beam Information
 - Mobility Events
 - Carrier Aggregation
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LTE-Advanced (LTE-A)

Provides measurements related to:

- Carrier Aggregation
 - High-Speed Data Transfer
 - Multiple Frequency Bands
 - Advanced Scheduling
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LTE

Supports comprehensive LTE performance monitoring, including:

- Signal Strength
- Cell Information

- Throughput
 - Handover Analysis
 - Protocol Decoding
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WCDMA

Engineers can monitor:

- Signal Quality
 - Coverage
 - Soft Handover
 - Radio Parameters
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HSPA and HSPA+

Supports monitoring of high-speed packet data services, including throughput, latency, and radio quality.

EDGE and GPRS

Although older technologies, Sigma-ML continues to support EDGE and GPRS for legacy network environments, ensuring backward compatibility during network modernization projects.

Wi-Fi

Sigma-ML can also collect Wi-Fi-related information, including:

- SSID
- RSSI
- Channel Information
- Signal Quality
- Access Point Identification

This enables comparison between cellular and Wi-Fi performance in mixed network environments.

13. VoLTE Measurement and Voice Quality Analysis

Voice over LTE (VoLTE) is one of the most important capabilities supported by Sigma-ML. Since voice quality directly impacts customer satisfaction, the application provides comprehensive tools for monitoring, recording, and analyzing VoLTE services.

Automatic Voice Call Origination

The application can automatically initiate voice calls according to predefined schedules. This enables engineers to perform large-scale voice quality testing without manual intervention.

Automatic Incoming Call Handling

Sigma-ML can automatically receive incoming calls, allowing continuous testing of call setup, connectivity, and termination.

Voice Recording

Voice calls can be recorded for detailed post-analysis. Recorded conversations help engineers investigate audio quality issues and compare subjective listening experiences with objective network measurements.

Silence Detection

Sigma-ML automatically detects periods of silence or audio interruption during a call. This feature helps identify temporary voice muting caused by network impairments.

RTP and SIP Monitoring

The application captures Real-time Transport Protocol (RTP) and Session Initiation Protocol (SIP) information, enabling engineers to analyze signaling procedures and media transmission quality.

POLQA MOS Measurement

Sigma-ML supports **Perceptual Objective Listening Quality Analysis (POLQA)** to calculate the **Mean Opinion Score (MOS)**. MOS provides an objective estimate of perceived voice quality on a numerical scale, helping engineers compare network performance across different locations and conditions.

Integrated Voice Quality Analysis

By combining voice recordings, protocol messages, radio measurements, and network statistics into a single dataset, Sigma-ML enables comprehensive root cause analysis of voice quality issues. Engineers can correlate call quality degradation with changes in signal strength, network handovers, or signaling events, leading to faster diagnosis and resolution of network problems.

14. Data Collection and Logging

One of the core strengths of Sigma-ML is its ability to collect comprehensive wireless network information while minimizing user intervention. During every measurement session, the application records radio parameters, protocol messages, network performance metrics, device status, and

location information. These logs provide engineers with a complete picture of network behavior and can be analyzed to identify performance issues or verify service quality.

The logging process begins as soon as a measurement session is initiated. Depending on the configured test profile, Sigma-ML continuously gathers data from the smartphone modem, operating system, and active network interfaces. Information is recorded at regular intervals to ensure that changes in network conditions are accurately captured throughout the testing period.

The application stores all collected information locally on the smartphone, allowing testing to continue even in areas with limited network connectivity. Once the test is complete, the recorded logs can be exported for detailed analysis using external reporting and visualization tools.

Information Recorded During Logging

Sigma-ML records a wide range of measurement parameters, including:

- Signal Strength (RSSI, RSRP, RSRQ, SINR)
- Cell Identifier (Cell ID)
- Physical Cell Identity (PCI)
- Frequency Band
- Network Technology (5G, LTE, WCDMA, etc.)
- GPS Coordinates
- Timestamp
- Download and Upload Throughput
- Latency and Round Trip Time
- Packet Loss
- Voice Call Status
- SMS Status
- Battery Level
- Device Storage Status
- Protocol Messages (Layer 3)
- Wi-Fi Parameters (SSID, RSSI)

The availability of these parameters allows engineers to correlate multiple network conditions simultaneously during troubleshooting.

15. Data Storage and Export

Efficient data management is essential for post-processing and reporting. Sigma-ML provides flexible storage options and supports industry-standard export formats.

Internal Storage

Measurement data can be stored directly in the smartphone's internal memory. This option is suitable for routine testing and short measurement sessions.

External Memory Card

For long-duration measurements or large datasets, Sigma-ML supports saving logs to external memory cards, reducing the risk of storage limitations during extended field operations.

Supported Export Formats

After completing a test, engineers can export measurement logs in the following formats:

- **CSV (Comma-Separated Values):** Easily imported into spreadsheet applications such as Microsoft Excel or analytical tools like Python and MATLAB for custom analysis and reporting.
- **DLF (Drive Log Format):** Preserves detailed measurement information for specialized network analysis applications and advanced troubleshooting workflows.

These export options ensure compatibility with a variety of reporting and visualization tools used by telecom operators and engineering teams.

16. Scheduled and Remote Testing

Large-scale network deployments often require measurements to be collected over extended periods or at specific times of the day. Sigma-ML addresses this requirement through scheduling and remote testing capabilities.

Scheduled Testing

Users can configure test scenarios to start automatically at predefined dates and times. This is particularly useful for:

- Peak-hour traffic analysis
- Night-time network monitoring
- Continuous performance evaluation
- Long-duration benchmarking
- Acceptance testing

By automating the test schedule, engineers can ensure that measurements are collected consistently without requiring on-site supervision.

Unattended Operation

Once a test profile has been configured, Sigma-ML performs all assigned tasks automatically. The application can continue operating for long periods while recording measurements and executing predefined test sequences.

Typical unattended activities include:

- Automatic FTP Download
- Automatic FTP Upload

- HTTP Download
- Voice Call Testing
- SMS Testing
- Ping Measurements
- Data Logging
- GPS Tracking

This capability enables organizations to perform **24 × 7 × 365** monitoring at remote locations with minimal human intervention.

17. Battery and Storage Monitoring

Since Sigma-ML operates on smartphones, efficient resource management is essential during extended measurement campaigns. The application continuously monitors device resources to prevent interruptions caused by insufficient battery power or storage capacity.

Battery Monitoring

Sigma-ML tracks the current battery level throughout the testing session. Engineers can estimate the remaining operating time and plan charging intervals accordingly.

Benefits include:

- Prevention of unexpected shutdowns
- Improved planning for long-duration tests
- Increased reliability of unattended measurements

Storage Monitoring

The application also monitors available storage space on the device. Before initiating long measurement sessions, users can verify that sufficient storage is available to save all generated logs.

This helps prevent:

- Incomplete log files
 - Data loss
 - Measurement interruptions
 - Export failures
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18. Real-Time Visualization

During active measurements, Sigma-ML provides multiple visualization options that allow engineers to monitor network performance in real time.

Chart View

Graphical charts display variations in parameters such as:

- Signal Strength
- Throughput
- Latency
- Voice Quality
- Network Technology Changes

These visualizations help engineers quickly identify trends, coverage gaps, and performance fluctuations.

Parameter View

Detailed numerical values for radio and protocol measurements are displayed in structured tables, enabling precise observation of network behavior.

Map View

Outdoor measurements can be visualized on a map using GPS coordinates, allowing engineers to correlate network performance with geographical location.

Event Timeline

Important events such as call drops, handovers, SMS failures, or user-defined markers are displayed chronologically, simplifying post-test analysis.

19. Network Optimization Workflow

Sigma-ML supports the complete lifecycle of wireless network optimization. The following workflow illustrates how engineers typically use the application during optimization projects.

Step 1 – Planning

Engineers define the measurement objectives, select the required tests, and configure the test profile.

Step 2 – Field Testing

The smartphone performs measurements while moving through the target area or remaining at a fixed location.

Step 3 – Data Collection

Network measurements are continuously recorded, including radio parameters, protocol messages, application performance, and GPS information.

Step 4 – Log Export

After testing, the collected data is exported in CSV or DLF format for detailed analysis.

Step 5 – Data Analysis

Engineers review the collected information to identify:

- Coverage gaps
- Weak signal areas
- Call drops
- High latency
- Low throughput
- Failed handovers
- Protocol errors

Step 6 – Network Optimization

Based on the analysis, optimization teams adjust network parameters such as antenna orientation, transmit power, neighbor cell configuration, and handover settings.

Step 7 – Verification

The measurement process is repeated to verify that the implemented changes have improved network performance.

20. Practical Applications

Sigma-ML is suitable for a wide range of wireless network evaluation and optimization scenarios.

Drive Testing

Engineers can perform vehicle-based measurements to evaluate network coverage, mobility performance, and handover behavior across large geographical areas.

Indoor Coverage Assessment

Shopping malls, office buildings, hospitals, airports, and educational institutions require reliable indoor wireless coverage. Sigma-ML supports indoor surveys using manual location markers when GPS is unavailable.

Network Acceptance Testing

Before a newly deployed network is commissioned, operators can use Sigma-ML to verify that coverage, throughput, and service quality meet contractual requirements.

Customer Complaint Investigation

When users report issues such as poor signal quality, slow internet speeds, or dropped calls, engineers can reproduce the problem using Sigma-ML and identify its root cause.

Network Expansion Projects

During the rollout of new base stations or additional frequency bands, Sigma-ML assists in validating the effectiveness of the deployment.

Performance Benchmarking

Operators can compare the performance of different network technologies, geographic regions, or service providers using standardized measurement profiles.

21. Business Benefits

Implementing Sigma-ML provides several operational and economic advantages for organizations involved in wireless network deployment and maintenance.

Reduced Operational Costs

Since the application operates directly on smartphones, organizations can reduce investment in expensive drive-test hardware and associated maintenance costs.

Improved Productivity

Automation of repetitive tasks allows engineers to focus on data analysis and network optimization rather than manual test execution.

Faster Problem Resolution

The availability of comprehensive radio, protocol, and application-layer information enables rapid identification of network issues and reduces troubleshooting time.

Increased Measurement Accuracy

Automated testing minimizes human error and ensures consistent measurement procedures across different projects and engineering teams.

Enhanced Customer Experience

By identifying and resolving network performance issues more quickly, operators can improve service quality, reduce call drops, and deliver better user experiences.

Scalable Deployment

Multiple smartphones equipped with Sigma-ML can be deployed simultaneously across different locations, allowing large-scale measurement campaigns without significant infrastructure investment.
